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Simulation-based Performance Analysis of an NR-U and WiFi Coexistence System in the Presence of Hidden Nodes

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***Abstract*—This paper analyzes the performance of an NR-U and WiFi coexistence system in the presence of hidden nodes based on simulation experiments. We consider a coexistence system where NR-U gNBs employ an NR-U category-4 LBT procedure while WiFi Access Points (APs) employ a WiFi DCF procedure for channel contention. The number of hidden nodes and that of sensible nodes of an NR-U gNB in the coexistence system are analyzed, and the impacts of major parameters on the system performance in terms of the average throughput and packet delay of an NR-U gNB/a WiFi AP are investigated through extensive simulation results. Useful results are obtained for the design of an NR-U and WiFi coexistence system.**

Index Terms*—*NR-U; WiFi; coexistence system; hidden node; performance analysis

I. Introduction

To address the ever-increasing demand for spectrum resources in cellular networks, 3GPP has specified New Radio (NR) based access to unlicensed spectrum (NR-U) to enable 5G network operation on unlicensed bands [1][2]. One of the most critical issues for a cellular network to operate on an unlicensed band is to ensure a fair and harmonious coexistence with other unlicensed systems, such as WiFi. For a fair coexistence, an NR-U system needs to introduce new functionalities, such as listen before talk (LBT) and discontinuous transmission (DTX) with limited maximum transmission duration[3], to enable an NR-U gNB to perform a clear channel assessment (CCA) check to sense the status of a channel before data transmission. In an NR-U and WiFi coexistence system, however, an NR-U gNB or WiFi access point (AP) unavoidably suffers from the hidden-node problem. For example, when a WiFi AP transmits a data packet to a user, if an NR-U gNB cannot detect the transmission because of its limited sensing range, the NR-U gNB will sense that the channel is idle and then transmit a data packet. In this case, the user may simultaneously receive the transmission from the WiFi AP and that from the NR-U gNB. As a result, a collision will occur, which would largely affect the performance of the coexistence system. Accordingly, the hidden-node problem should be taken into account in the performance analysis of an NR-U and WiFi coexistence system. However, most existing relevant work did not consider the impact of hidden nodes[4-7]. There are only limited work taking the hidden-node problem into account. In [8], Lee et al. analyzed the performance of an LAA and WLAN coexistence system with asymmetric hidden terminals based on two joint Markov chains. In [9], Zhang et al. proposed a hidden-node-aware joint licensed and unlicensed resource allocation algorithm for an LAA system. In [10], Lee and Yang proposed a downlink interference control scheme for an LAA-LTE eNB to prevent the asymmetric hidden WiFi APs. In [11], Sathya et al. performed extensive experiments on the campus of the University of Chicago, which presents a perfect hidden node scenario where WiFi APs and a LAA base station are hidden from each other. It was found that the performance of the WiFi APs is affected by the hidden node problem more severely than that of the LAA base station. Therefore, the performance analysis of an NR-U and WiFi coexistence system in the presence of hidden nodes is worth further studying for investigating the impacts of relevant parameters on the system performance.

In this paper, we analyze the performance of an NR-U and WiFi coexistence system in the presence of hidden nodes based on simulation experiments. We consider a coexistence system where NR-U gNBs employ a NR-U category-4 LBT procedure while WiFi APs employ a WiFi DCF procedure for channel access. We first analyze the number of hidden nodes and that of sensible nodes of an NR-U gNB and then investigate the impacts of major parameters on the system performance in terms of the average throughput and packet delay of an NR-U gNB/a WiFi AP through extensive simulation results. This work reveals several useful results for the design of an NR-U and WiFi coexistence system.

The remainder of this paper is organized as follows. Section II describes the system model considered in this paper and introduces the NR-U category-4 LBT procedure. Section III analyzes the mean number of hidden nodes and that of sensible nodes of an NR-U gNB. Section IV investigates the impacts of relevant parameters on the system performance through simulation results. Section VI concludes this paper.

II. System Model

In this section, we describe the system model considered in this paper and introduce the NR-U category-4 LBT procedure.

A. System model

We consider the downlink data transmission in an NR-U and WiFi coexistence system operating on an unlicensed spectrum band. The coexistence system consists of NR-U gNBs, WiFi APs, and a number of NR-U users and WiFi users.

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The NR-U gNBs employ the NR-U category-4 LBT procedure while the WiFi APs employ the DCF procedure with an RTS/CTS mechanism to contend for the channel.

In the coexistence system, the spatial distribution of NR-U gNBs follows a homogeneous Poisson point process (PPP) with density υ_u and that of WiFi APs follows a homogeneous PPP with density υ_w. The sensing distance of an NR-U gNB is $r_{u,s}$ and the transmission distance of an NR-U gNB is $r_{u,t}$. The sensing distance of a WiFI AP is $r_{w,s}$ and the transmission distance of a WiFi AP is $r_{w,t}$. The data packet arrival of an NR-U gNB follows a Poisson process with a mean rate λ_u and that of a WiFi AP follows a Poisson process with a mean rate λ_w.

B. NR-U category-4 LBT procedure

In the coexistence system considered, the NR-U gNBs employ the NR-U category-4 LBT procedure to perform channel contention for data transmission. When an NR-U gNB has a data packet to transmit, it will first perform channel sensing. If the channel is sensed idle for a defer duration (T_d), the NR-U gNB will enter a backoff process and start with an initial backoff stage. A backoff counter will be started with its initial value randomly chosen from $[0, W_{u,0}-1]$, where $W_{u,0}$ is the initial contention window (CW) size of an NR-U gNB. If the NR-U gNB senses the channel idle for a CCA slot (T_s), the value of the backoff counter will be decremented by 1. If the channel is sensed busy, the NR-U gNB will first freeze the backoff counter and then continue to perform channel sensing. If the channel is sensed idle for a defer duration T_d again, the NR-U gNB will recover the backoff counter and continue the backoff process. When the backoff counter reaches 0, the NR-U gNB will transmit the data packet immediately. If the transmission is successful, the NR-U gNB will wait for a new data packet to transmit. If the data packet is collided, the NR-U gNB will enter the next backoff stage (i) to retransmit the data packet. At backoff stage i, the backoff counter is reset to a value randomly chosen from $[0, W_{u,i}-1]$, where $W_{u,i}$ is the contention window (CW) size of an NR-U gNB at backoff stage i. The value of the contention window (CW) size is given by

$$W_{u,i} = \begin{cases} 2^i W_{u,0} & 0 \le i \le m_u \\ 2^{m_u} W_{u,0} & m_u < i \le m_u + h_u \end{cases}, \tag{1}$$

where m_u is the backoff order corresponding to the maximum contention widow size for an NR-U gNB and h_u is the limit of the number of retransmissions for an NR-U gNB after the backoff order reaches m_u and beyond. If the data packet is still collided at the maximum backoff stage $m_u + h_u$, the NR-U gNB will discard the data packet and wait for a new data packet to transmit. Due to the limitation of space, readers are referred to [3] for more details on the NR-U categaory-4 LBT procedure.

III. Analysis of the Number of Sensible Nodes and that of Hidden Nodes

In this section, we first analyze the mean number of sensible nodes and that of hidden nodes of an NR-U gNB, and then show the impact of the sensing distance of an NR-U gNB.

A. The mean number of sensible nodes and that of hidden nodes of an NR-U gNB

As shown in Fig. 1, when a source node (an NR-U gNB or a WiFi AP) contends for a channel to transmit a data packet to a destination user, if the source node can sense the transmission from a node (an NR-U gNB or a WiFi AP), the node is referred to as a sensible node (S_node). If a node cannot sense the transmission from the source node to the destination user but the destination user can receive the transmission from the node, the node is a hidden node (H_node).

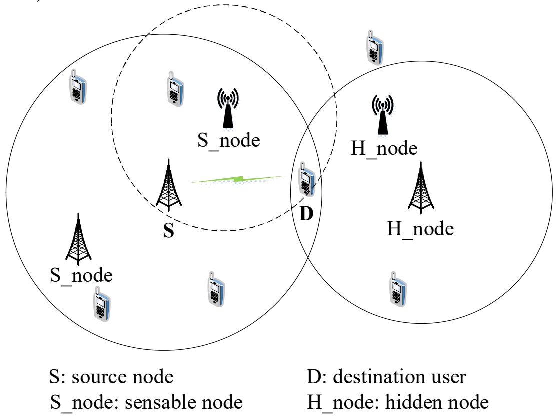

Fig. 1. Illustration of hidden nodes and sensible nodes.

1) The mean number of sensible NR-U gNBs and that of hidden NR-U gNBs of an NR-U gNB

Consider a typical NR-U gNB A which transmits a data packet to an NR-U user B, as shown in Fig. 2. The distance between the typical gNB A and the NR-U user B is denoted by r. The shaded area is an area which is determined by the typical gNB A's transmission distance and sensing distance. C is the boundary point of the shaded area, which it is the intersection of the circle centered at A with radius $r_{u,s}$ and the circle centered at B with radius $r_{u,t}$. If an NR-U gNB is located in the shaded area, the NR-U gNB cannot sense the transmission from the typical gNB A, but the NR-U user B can simultaneously receive the transmission from the typical gNB A and that from the NR-U gNB. Thus, the NR-U gNB is a hidden node of the typical gNB A. The shaded area is the area in which all hidden NR-U gNBs of the typical gNB A are located, called hidden NR-U gNB area for an NR-U gNB.

Let $\theta_{u,u}$ denote the angle between AC and AB in Fig. 2. Obviously, we have

$$\theta_{u,u} = \arccos \frac{(r_{u,s})^2 + r^2 - (r_{u,t})^2}{2 r_{u,s} r}. \tag{2}$$

Let S^u_{uh} denote the area of the hidden NR-U gNB area for the typical gNB A in Fig.2. According to geometry, S^u_{uh} is given by Eq.(3).

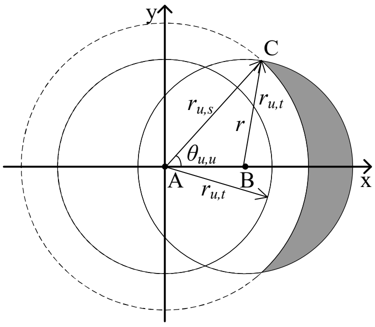

Fig. 2 The hidden NR-U gNB area for an NR-U gNB.

Since an NR-U user is randomly located in the transmission range of an NR-U gNB, the mean area of the hidden NR-U gNB area for an NR-U gNB is given by

$$\overline{S^u_{uh}} = \int_{r_{u,s}-r_{u,t}}^{r_{u,t}} \frac{2\pi r}{\pi (r_{u,t})^2} S^u_{uh} dr \,. \tag{4}$$

Therefore, the mean number of hidden NR-U gNBs for an NR-U gNB is given by

$$n^u_{uh} = \upsilon_u \overline{S^u_{uh}} \,. \tag{5}$$

The mean number of sensible NR-U gNBs for an NR-U gNB is given by

$$n^u_{us} = \upsilon_u \pi (r_{u,s})^2 \,. \tag{6}$$

2) The mean number of sensible WiFi APs and that of hidden WiFi APs of an NR-U gNB

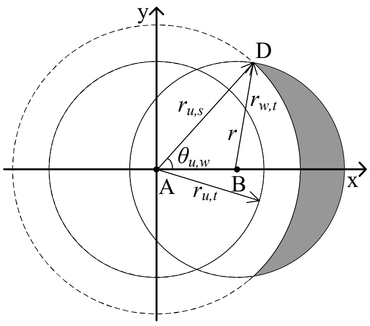

Fig. 3 The hidden WiFi AP area for an NR-U gNB.

Consider a typical NR-U gNB A which transmits a data packet to an NR-U user B, as shown in Fig. 3. The distance between the typical gNB A and the NR-U user B is denoted by r. D is the boundary point of the shaded area, which is the intersection of the circle centered at A with radius $r_{u,s}$ and the circle centered at B with radius $r_{w,t}$. If a WiFi AP is located in the shaded area, the WiFi AP cannot sense the transmission from the typical gNB A, but the NR-U user B can simultaneously receive the transmission from the typical gNB A and that from the WiFi AP. Thus, the WiFi AP is a hidden node of the typical gNB A. The shaded area is the area in which all hidden WiFi APs of the typical gNB A are located, called hidden WiFi AP area for an NR-U gNB.

Let $\theta_{u,w}$ denote the angle between AD and AB in Fig. 3. Obviously, we have

$$\theta_{u,w} = \arccos \frac{(r_{u,s})^2 + r^2 - (r_{w,t})^2}{2 r_{u,s} r} \,. \tag{7}$$

Let S^u_{wh} denote the area of the hidden WiFi AP area for the typical gNB A in Fig. 3. According to geometry, S^u_{wh} is given by Eq.(8). Thus, the mean area of the hidden WiFi AP area for an NR-U gNB is given by

$$\overline{S^u_{wh}} = \int_{r_{u,s}-r_{w,t}}^{r_{u,t}} \frac{2\pi r}{\pi (r_{u,t})^2} S^u_{wh} dr \,. \tag{9}$$

The mean number of hidden WiFi APs for an NR-U gNB is given by

$$n^u_{wh} = \upsilon_w \overline{S^u_{wh}} \,. \tag{10}$$

The mean number of sensible WiFi APs for an NR-U gNB is given by

$$n^u_{ws} = \upsilon_w \pi (r_{u,s})^2 \,. \tag{11}$$

B. Impact of the sensing distance of an NR-U gNB on the number of sensible nodes and that of hidden nodes of an NR-U gNB

Fig. 4 shows the impact of the sensing distance of an NR-U gNB ($r_{u,s}$) on the number of sensible nodes (n^u_{us} , n^u_{ws}) and that of hidden nodes (n^u_{uh} , n^u_{wh}) for an NR-U gNB, where $\upsilon_u = \upsilon_w = 0.0001/m^2$, $r_{w,s} = 90m$, $r_{w,t} = 70m$, $r_{u,t} = 80m$. It is seen that as the sensing distance of an NR-U gNB increases, the number of hidden WiFi APs and that of NR-U gNBs decrease while the number of sensible NR-U gNBs and that of sensible WiFi APs increase. When the sensing distance of an NR-U gNB is larger than twice the transmission distance of an NR-U gNB, the number of the hidden NR-U gNBs becomes 0, but the number of sensible NR-U gNBs and that of sensible WiFi APs are large.

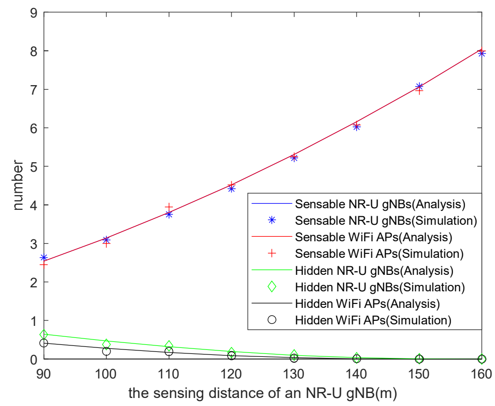

Fig. 4 The impact of the sensing distance of an NR-U gNB.

$$S^u_{uh} = 2\times\left[\int_{r_{u,s}\cos\theta_{u,u}}^{r_{u,s}} \left(\sqrt{(r_{u,t})^2-(x-r)^2} - \sqrt{(r_{u,s})^2-x^2}\right)dx + \int_{r_{u,s}}^{r_{u,t}+r} \sqrt{(r_{u,t})^2-(x-r)^2}\,dx\right] \tag{3}$$

$$S^u_{wh} = 2\times\left[\int_{r_{u,s}\cos\theta_{u,w}}^{r_{u,s}} \left(\sqrt{(r_{w,t})^2-(x-r)^2} - \sqrt{(r_{u,s})^2-x^2}\right)dx + \int_{r_{u,s}}^{r_{w,t}+r} \sqrt{(r_{w,t})^2-(x-r)^2}\,dx\right] \tag{8}$$

IV. Simulation Results

In this section, we analyze the performance of the coexistence system through simulation results. The simulation experiments were conducted on a discrete-event driven simulator which we built using Matlab. We use the throughput and packet delay of an NR-U gNB/a WiFi AP as the performance metrics. The throughput of an NR-U gNB/a WiFi AP is defined as the average number of data bits that are successfully transmitted by an NR-U gNB/a WiFi AP per unit time. The packet delay of an NR-U gNB/a WiFi AP is defined as the time from the instant an NR-U gNB/a WiFi AP starts to contend for the channel to transmit a data packet to the instant the data packet is successfully transmitted. The parameter values used in the simulation experiments are given in Table I.

TABLE I. System Parameters[12]

The length of a timeslot (T_s)	9μs
The density of WiFi APs (υ_w)	0.0001 APs/m^2
The density of NR-U gNBs (υ_n)	0.0001 gNBs/m^2
The sensing distance of an WiFi AP ($r_{w,s}$)	90m
The transmission distance of an WiFi AP ($r_{w,t}$)	70m
The sensing distance of an NR-U gNB ($r_{u,s}$)	100m
The transmission distance of an NR-U gNB ($r_{u,t}$)	80m
WiFi data packet arrival rate (λ_w)	3 packets/ms
NR-U data packet arrival rate (λ_n)	3 packets/ms
The payload size of a WiFi packet	12000bits
The size of a WiFi MAC header	192bits
The size of a WiFi PHY header	224bits
The size of a WiFi RTS packet	160bits+PHY header size
The size of a WiFi CTS packet	112bits+PHY header size
The size of a WiFi ACK packet	112bits+PHY header size
The channel transmission rate of a WiFi AP(R_w)	54Mbps
WiFi SIFS (T_{SIFS})	16μs
WiFi DIFS (T_{DIFS})	34μs
WiFi initial CW size ($W_{w,0}$)	16
The number of WiFi CW stages (m_w)	6
The maximum number of WiFi retransmissions (m_w+h_w)	9
The payload size of an NR-U packet	12000bits
The size of an NR-U ACK packet	300bits
NR-U defer period (T_d)	34μs
The channel transmission rate of an NR-U gNB (R_u)	100Mbps
NR-U initial CW size ($W_{u,0}$)	16
The number of NR-U CW stages (m_u)	6
The maximum number of NR-U retransmissions (m_u+h_u)	9

A. The impact of the data packet arrival rate

Fig. 5 shows the average throughput of an NR-U gNB and that of a WiFi AP with the data packet arrival rate of NR-U gNBs or WiFi APs, respectively, where we assume that the data packet arrival rate of an NR-U gNB is equal to that of a WiFi AP. It is seen that with the data packet arrival rate increasing, the average throughput of a WiFi AP increases dramatically first and then tends to be relatively stable. The average throughput of an NR-U gNB increases dramatically first, and then gradually decreases and tends to be relatively stable. When the data packet arrival rate is small, the average throughput of a WiFi AP and that of an NR-U gNB are basically equal. This is because in this case all APs and gNBs are in an unsaturated state, and the number of APs or gNBs that have a data packet to transmit is small. As a result, an AP or gNB that has a data packet to transmit can quickly transmit the data packet, and the collision possibility of the data packet is small. When the data packet arrival rate is large, the average throughput of a WiFi AP and that of a NR-U gNB tend to be relatively stable. This is because in this case all APs and gNBs are in a saturated state. Meanwhile, the average throughput of a WiFi AP is larger than that of an NR-U gNB. This is because the sensing distance of a WiFi AP is smaller than that of a NR-U gNB, as given in Table I, and thus the sensible nodes of a WiFi AP is smaller than that of an NR-U gNB.

Fig. 6 shows the average packet delay of an NR-U gNB and that of a WiFi AP with the data packet arrival rate. It is seen that with the data packet arrival rate increasing, both the average packet delay of an NR-U gNB and that of a WiFi AP increase first and then tend to be relatively stable. When the data packet arrival rate is small, the average packet delay of an NR-U gNB is smaller than that of a WiFi AP. This is because that the channel transmission rate of an NR-U gNB is larger than that of a WiFi AP, as given in Table I, and thus the transmission time of an NR-U data packet is smaller than that of a WiFi data packet. When the data packet arrival rate is beyond a certain value, the average packet delay of an NR-U gNB becomes larger than that of a WiFi AP.

B. The impact of the density of NR-U gNBs

Fig. 7 shows the average throughput of an NR-U gNB and that of a WiFi AP with the density of NR-U gNBs. It is seen that with the density of NR-U gNBs increasing, both the average throughput of an NR-U gNB and that of a WiFi AP decrease. This is because with the number of NR-U gNBs increasing, the probability that a node transmits a data packet decreases but the collision possibility of a data packet increases.

Fig. 8 shows the average packet delay of an NR-U gNB and that of a WiFi AP with the density of NR-U gNBs. It is seen that with the density of NR-U gNBs increasing, both the average packet delay of an NR-U gNB and that of a WiFi AP increase.

C. The impact of the sensing distance of an NR-U gNB

Fig. 9 shows that the average throughput of an NR-U gNB and that of a WiFi AP with the sensing distance of an NR-U gNB. It is seen that with the sensing distance of an NR-U gNB increasing, the average throughput of an NR-U gNB decreases while that of a WiFi AP increases. This is because with the sensing distance of an NR-U gNB increasing, although the number of hidden nodes of an NR-U gNB decreases, the number of sensible nodes of an NR-U gNB increases. Thus, the probability that an NR-U gNB transmits a data packet decreases and thus the probability that a WiFi AP transmits a data packet increases.

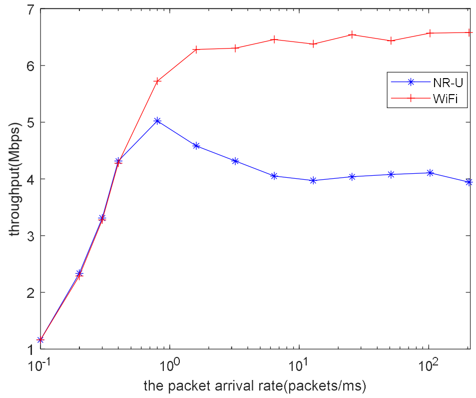

Fig. 5 Throughput vs the packet arrival rate.

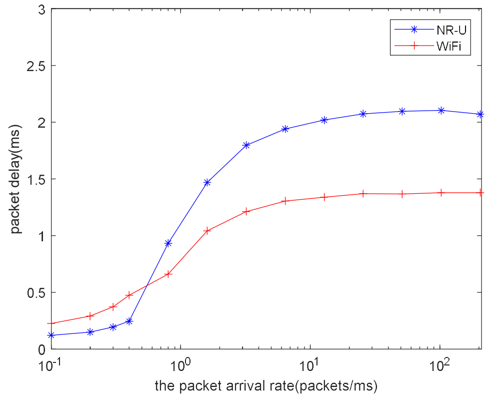

Fig. 6 Packet delay vs the packet arrival rate.

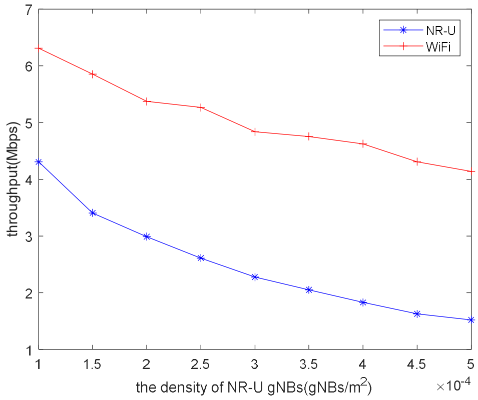

Fig. 7 Throughput vs the density of NR-U gNBs.

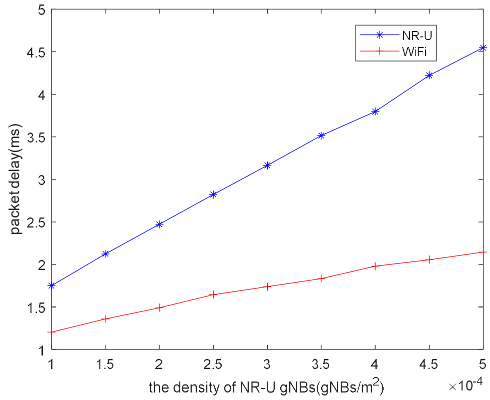

Fig. 8 Packet delay vs the density of NR-U gNBs.

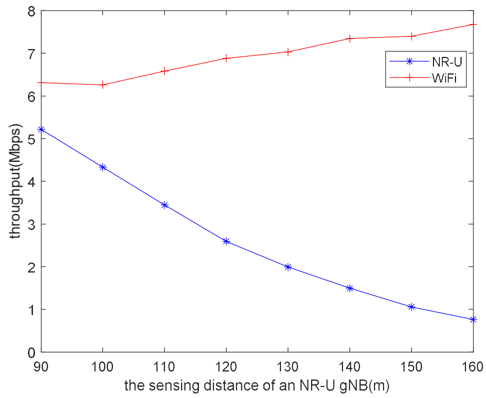

Fig. 9 Throughput vs the sensing distance of an NR-U gNB.

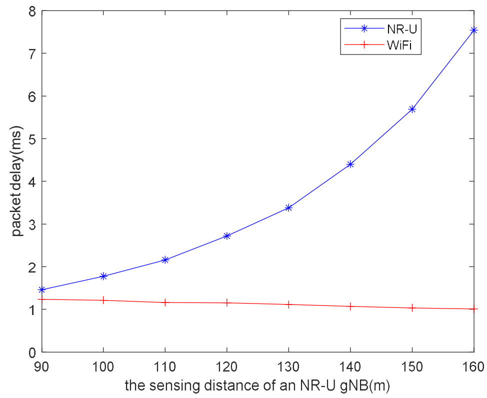

Fig. 10 Packet delay vs the sensing distance of an NR-U gNB.

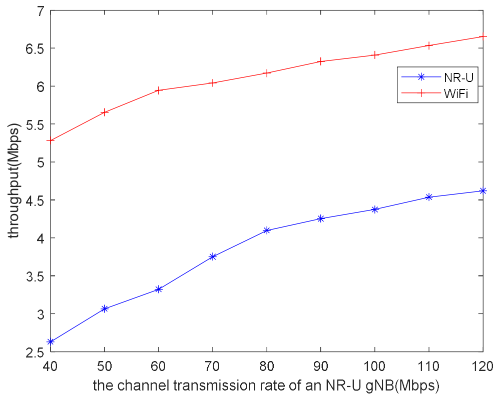

Fig. 11 Throughput vs the channel transmission rate of an NR-U gNB.

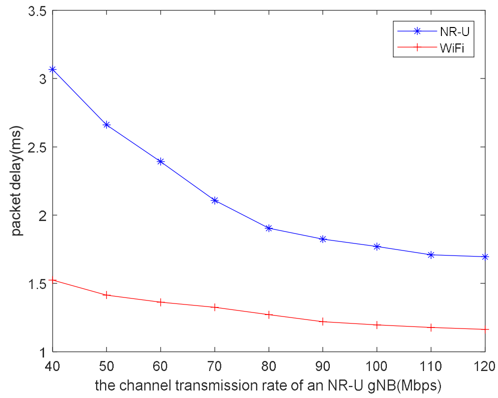

Fig. 12 Packet delay vs the channel transmission rate of an NR-U gNB.

Fig. 10 shows that the average packet delay of an NR-U gNB and that of a WiFi AP with the sensing distance of an NR-U gNB. It is seen that with the sensing distance of an NR-U gNB increasing, the average packet delay of an NR-U gNB increases remarkably while the average packet delay of a WiFi AP decreases slightly.

D. The impact of the channel transmission rate of an NR-U gNB

Fig. 11 shows that the average throughput of an NR-U gNB and that of a WiFi AP with the channel transmission rate of an NR-U gNB. It is seen that with the channel transmission rate increasing, both the average throughput of an NR-U gNB and that of a WiFi AP increase. This is because with the channel transmission rate increasing, the transmission time of an NR-U data packet decreases, and thus an NR-U gNB or WiFi AP can transmit more data packets per unit time.

Fig. 12 shows that the average packet delay of an NR-U gNB and that of a WiFi AP with the channel transmission rate of an NR-U gNB. It is seen that with the channel transmission rate increasing, both the average packet delay of an NR-U gNB and that of a WiFi AP decrease.

V. Conclusion

In this paper, we analyzed the performance of an NR-U and WiFi coexistence system in the presence of hidden nodes based on simulation experiments. In the coexistence system, the NR-U gNBs employ a NR-U category-4 LBT procedure while WiFi APs employ a WiFi DCF procedure for channel access. We first analyzed the number of hidden nodes and that of sensible nodes of an NR-U gNB, and then investigated through extensive simulation results the impacts of major parameters on the throughput and average packet delay of an NR-U gNB/a WiFi AP, including the arrival rate of data packets, the density of NR-U gNBs, the sensing distance of an NR-U gNB, and the channel transmission rate of an NR-U gNB. The simulation results reveal that a higher density of NR-U gNBs would degrade the performance of both the NR-U system and the WiFi system. A larger sensing distance of an NR-U gNB would degrade the performance of the NR-U system but is beneficial to the performance of the WiFi system. In addition, a larger channel transmission rate of an NR-U gNB is beneficial to the performance of both the NR-U system and the WiFi system.

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